Extending S_λ below zero: the recursion $S_\lambda = \frac{\sqrt{\beta}}{2\lambda} b S_{\lambda+1/2}$

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Abstract

Unit 19 of the grammar-paper §4 autoformalisation: the extension of the fluctuation function to non-positive order (`defn:S_negative`). For $\lambda \le 0$ (away from the poles $\{0, -\frac{1}{2}, -1, \dots\}$) the paper defines $S_\lambda = \frac{\sqrt{\beta}}{2\lambda} b S_{\lambda+1/2}$, lowering the order until the integral regime is reached. We formalise this recursion and prove the *lowering* ladder relation of `lem:extended_ladder`, which is definitional. Zero `sorry`/`axiom`.

1 Setting

The closed form $S_\lambda = 2^{1-\lambda}\beta^{-\lambda}\Gamma(2\lambda)e^{\beta a^2/8}D_{-2\lambda}(-a\sqrt{\beta/2})$ continues meromorphically in λ, but our `parCylNeg` is the positive-order integral representation only. The paper's own extension is instead recursive, via the lowering operator b from the ladder algebra (unit 16). This is the final structural piece of §4.

2 The things formalised

```
noncomputable def Sneg (    :  ) :   →   →
  | 0       => fun a => fluctuation    a
  | (k + 1) => fun a =>
       ^ ((1: )/2) / (2 * (  - (↑(k+1))/2)) * lowerOp   (Sneg    k) a

theorem Sneg_lowering (    :  ) (h  : 0 <  ) (k :  ) (hlam :   - (↑(k+1))/2   0) (a :  ) :
    lowerOp   (fun a => Sneg    k a) a
      = (2 * (  - ↑k/2) - 1) *   ^ (-(1: )/2) * Sneg    (k + 1) a
```

‘Sneg   k‘ represents $S_{\mu-k/2}$, indexing the extension by the number of half-steps taken below a base order $\mu > 0$.

3 Proof strategy

The recursion is structural on k (Lean accepts it directly; `lowerOp` uses `deriv`, which is total, so no differentiability obligation enters the definition). The lowering relation is the definition rearranged: $S_{\rho-1/2} = \frac{\sqrt{\beta}}{2\rho'} b S_\rho$ with $\rho' = \rho - \frac{1}{2}$, so $bS_\rho = \frac{2\rho'}{\sqrt{\beta}} S_{\rho-1/2} = (2\rho-1)\beta^{-1/2}S_{\rho-1/2}$. In Lean: rewrite S_{k+1} by its defining equation, group the scalar prefactor as $\frac{2\rho-1}{2\rho'} \cdot \beta^{-1/2}\beta^{1/2}$, and collapse it with $\beta^{-1/2}\beta^{1/2} = 1$ and $2\rho - 1 = 2\rho'$ (`div_self`, needing $\rho' \neq 0$ — the non-pole hypothesis).

4 Roadblocks and resolutions

Almost none: a small scalar-cancellation. Two points of care — the numeric identity $2(\mu - \frac{k}{2}) - 1 = 2(\mu - \frac{k+1}{2})$ is by `push_cast; ring` (the $\uparrow(k+1)$ cast), and the non-pole hypothesis $\mu - \frac{k+1}{2} \neq 0$ is exactly what makes the prefactor invertible (`div_self`). (Recall `h`λ does not parse — λ is the anonymous-function token; use `hlam`.)

5 What was Mathlib, what was new

Only `Real.rpow_add/div_self` and `push_cast` from Mathlib. New is the recursive extension itself and the observation that the extended lowering relation is definitional — no analysis needed once the ladder operator `lowerOp` (unit 16) is in hand.

6 Lessons

When a special function is extended by an operator recursion, the ladder relation *in the direction of the recursion* is free (rearrange the defining equation); it is the *opposite* direction (raising) that carries the analytic content, requiring smoothness of the extension and the Weyl commutator.

7 Follow-ups

The remaining half of `lem:extended_ladder` — the raising relation $b^\dagger S_\lambda = \beta^{1/2} S_{\lambda+1/2}$ for the extension — and the general-Q number operator $N_\lambda S_{\lambda+Q/2} = Q S_{\lambda+Q/2}$ both require the C^∞-smoothness of `Sneg` (to apply $b, b^\dagger$ and `ladder_commutator`), proved by induction from the smoothness of the integral S_μ and smoothness-preservation under `lowerOp`. With this, the entire ladder/log structure of grammar §4 is formalised.